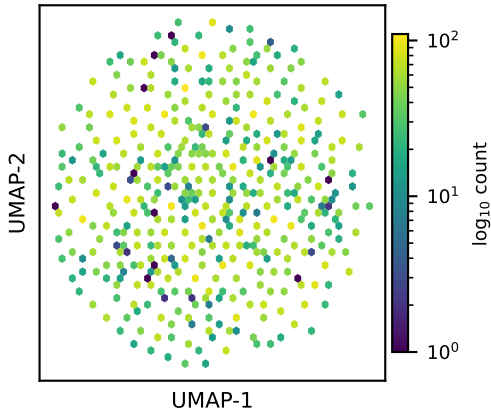
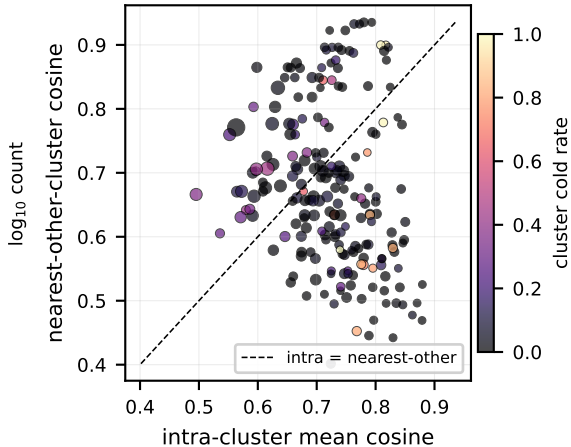


(A) ECAPA UMAP manifold
(19,101 chunks, log density)



(B) Per-cluster tightness vs
separation (k=210, marker $\propto$ size)



(C) raw-L2 192-D validation

Validated: SHIPPED k=210 labels
(production cluster.py space;
NOT the optimistic UMAP-32)

kNN label cohesion (k=10)
0.694 vs chance 0.0074
= 94x (load-bearing)

HDBSCAN (independent)
406 clusters, noise 3.8%
non-degenerate; NMI 0.77
(ARI 0.31: 406>210 granularity)

Agglomerative@210 NMI 0.76
Silhouette 0.073 (relative only)